WINXON NGUYEN
Los Angeles, CA | 626 727 1564 | winxonnguyen44@gmail.com | LinkedIn | Portfolio | GitHub

EDUCATION

University of California-Los Angeles, Los Angeles, CA June, 2028
- *BS in Applied Mathematics w/ Specialization in Computing | BS in Statistics & Data Science*
- **Relevant Coursework***: Abstract Linear Algebra, Honours Physics, Discrete Structures, Differential Equations, Real Analysis, Data Structures & Algorithms, Numerical Analysis, Machine Learning*

EXPERIENCE & PROJECTS

CruX Neuroscience, *Developer*, Los Angeles, CA September 2025-Present
- Built an end-to-end **ETL pipeline** in **Python** processing live 16-channel EEG data into a real-time focus-state classifier; framing the task as binary classification enabled threshold-based decision rules that reached 99% accuracy across experimental sessions.
- Applied signal-processing techniques to cut EEG artifact/noise by ~79%, improving classifier reliability across multiple users.
- Exposed the classifier through a **Python API** integrated into users' existing workflows, cutting time-on-task by 60-80% versus baseline completion times reported on comparable public datasets.

UCLA: Math + Code + Art, Los Angeles, CA April 2026-Present
Researcher
- Developed **Python** applications creating higher dimensional Voronoi tessellations in art and music. Built out visual **deep learning** model to measure art and music styles, and transfer art semantic across pieces.
- Deconstructed **Meta's Segment Anything Model (SAM) image encoder** with a faculty advisor to study its object-segmentation reasoning, then derived a custom Voronoi density-field equation that converts segmented images into multi-dimensional generative mosaics
- Researched pipeline for Image Segmentation, **Fourier Feature Mapping** and **Vision Transformer** process to optimize promptable segmentation.

Lead Developer
- Lead development of interactive app in **Python** for Oceanside Museum of Art. Allows users to turn any custom image into a Voronoi tessellation with variable number of centroidal nodes and image weights. Incorporated **Lloyd relaxation algorithm** to increase user control over their custom art piece.
- Integrated **Meta SAM image encoder** into the pipeline to allow users to prompt generalized image segments.
- Collaborated with team to create an algorithm for using photo collages to replace image pixels, allowing users to create mosaics of their memories, from public database images.

UCLA Startup Labs: Data Infrastructure for ML in Predictive Systems, *Founder,* Los Angeles, CA 2026
- Built a scraping and embedding pipeline combining a custom CNN (image) and an adapted **NLP model** (text) to map multi-modal social signals into a 4-dimensional semantic feature space.
- Backtested a downstream trading strategy on ~50 event-contract markets on Kalshi over a 4-month window, achieving a **Sharpe ratio of 1.38.**

C-VUSD AI Task Force, *Team Lead*, Walnut, CA August 2024 - June 2025
- Designed district-wide AI adoption framework for scalable use in education through policy and teaching psychology.
- Trained specialized **RNN agents** for personalized teaching and administered use. Distributed to 20,000+ users.

VoroNav: Offline Navigation System for Autonomous Systems, *Developer*, Los Angeles, CA 2026
- Built a from-scratch parallel Voronoi engine, Jump Flooding, with **NumPy**, multithreaded **Numba**, and **PyTorch** backend to produce a bit-identical output and checked against an exact solver (358 tests). Diagnosed slow 64-bit math on the Apple GPU and wrote a custom Metal kernel that sped it up 100×: a **2048² diagram in 14 ms**, about 300× faster than NumPy.
- Designed a maximum-clearance Voronoi roadmap planner and benchmarked it against A*, *clearance-weighted A** and RRT* on 94 maps (4,100 queries, 58K planner runs). Its paths were at least as far from walls as A**'s in 99.5% of queries, with 70% more average clearance. **It tied tuned weighted A* on safety with 1.5–3.6× faster queries**.
- Built an offline perception-to-planning pipeline (SAM 2 floor masks → bird's-eye grid → roadmap) and an error-injection study (190K graded plans) showing that clearance halves collisions from segmentation leaks but not from missed obstacles. On first-person video, collisions fell from 0.6–1.1% (A*) to ≤0.1%, and a leak-detection gate I designed raised recall from 8% to 65%.

TECHNICAL SKILLS

- *Computer Languages:* C++, C, Python, SQL, R
- *Tools:* Pandas, NumPy, PyTorch, Data Pipeline, Regression, Time Series Analysis, ETL, Vision Transformers, CNNs
- *Languages:* Fluent: English, Vietnamese, Mandarin
- *Awards:* Math Department (’22, ’23, ’25), C-VUSD Robotics Award ’17, CIF Athletic Scholar Award ’25, International Scholar ’25, SoCal Field Band (Gold 2021, Bronze ’22, Silver‘24), Louis Armstrong Award ’25